

An aerostructural reduced-order model

Identifying the LEI-V3 quasi-steady ROM from coupled simulations only

Model, identification method, uncertainty and validation

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Abstract

The quasi-steady reduced-order model (ROM) of Cayon, van Deursen and Schmehl [1] needs a lift and a drag polar, the bridle pitch θ_b between the bridle resultant and the wing chord, and a steering gain. The existing LEI-V3 ROM is semi-empirical: rigid-wing vortex-step-method (VSM) polars with drag offsets and a θ_b tuned to flight data. This note documents a second ROM in which *every* parameter is identified from coupled VSM + Billow-wireframe solutions at the centre of the wind window. The lift and drag polynomials are selected by forward-backward stepwise regression on a cross-validation error that holds out whole frozen shapes, include a smooth stall switch, and carry anchor-cluster bootstrap intervals. The ROM reproduces the coupled trims it was identified from to about one percent in speed and tension. Validating it against the 2019 flight exposed a frame error in the ROM code itself: the bridle pitch and roll were taken in a frame rotated by twice the sideslip angle off the apparent wind, which put the bridle angle of attack 2–3° off in turns (and zero off at the centre-window states the ROM was identified on). With that corrected, the simulation-only ROM reproduces held-out reel-out cycles to +7% in tension and +10% in speed with the measured turn rate as input, while the paper’s flight-calibrated ROM, which had absorbed the error, read –19% and –11% until it was recalibrated the same way. A third ROM keeps the simulated lift polynomial and stall and refits the bridle pitch $\theta_b(u_p)$, the roll gain and a single-input drag correction $\Delta C_D(u_p, u_s^2)$ to the 2019 flight, first on the EKF coefficients and then jointly by output error on the quasi-steady solve with the steering gain set by the turn-rate law; it reproduces the held-out reel-out to –3% in tension and –1% in speed. Reel-in remains unresolved by every ROM.

Contents

1	Purpose and scope	2
2	The model	2
2.1	Inputs and conventions	2
2.2	Lift and drag	3
2.3	Bridle pitch and steering	3
2.4	KCU drag, a separate force	3
3	Data	3
4	Identification method	4
4.1	Bridle pitch	4
4.2	Steering gain	4
4.3	Stall location	4
4.4	Candidate terms	4

4.5	Least squares and selection	4
4.6	Metrics and uncertainty	5
5	Results	5
5.1	Identified coefficients	5
5.2	Check against the coupled trims	7
5.3	Flight validation	8
6	The bridle-angle frame	12
7	Flight correction	12
7.1	Steps 1–2: equation error on the EKF coefficients	13
7.2	Step 3: output error on the quasi-steady solve	15
7.3	Result	16
8	Optimisations	17
9	Limitations and open points	18
10	Reproduction	20

1 Purpose and scope

The ROM is a point mass in the course reference frame whose wing is assumed to align instantaneously with the bridle resultant [1]. Its aerodynamic inputs are

- the wing lift and drag coefficients C_L , C_D ;
- the geometric bridle pitch θ_b that turns the bridle angle of attack α_b into the wing angle of attack, $\alpha_w = \alpha_b - \theta_b$ (paper Eqs. 1–2);
- the steering gain $k_{\phi,s}$ of the control-induced aerodynamic roll, $\phi_{a,w} = k_{\phi,s} u_s$ (paper Eq. 43).

The semi-empirical LEI-V3 ROM (`rom_config_semi_empirical.yaml`) takes its polars from rigid-wing VSM and calibrates drag offsets and θ_b on flight-identified states [1, Sect. 5.1]. The aerostuctural ROM (`rom_config_aerostuctural.yaml`) is identified from simulations only: the VSM aerodynamics of the *deformed* kite, coupled to the Billow structural model. It is the simulation-only reference against which flight data can then be used for a minimal correction.

Scope and limits, stated once:

- centre of the wind window (azimuth and elevation 0), gravity off, tetherless trims (the ROM models gravity and the tether itself);
- 2019 LEI-V3 hardware; power-tape band $u_p \in [1.6, 2.2]$ m; steering up to the coupled data's $|u_s| \leq 0.18$;
- the stall comes from VSM with artificial viscosity on frozen shapes.

2 The model

2.1 Inputs and conventions

- u_p is the power-tape length l_{dp} in metres.
- u_s is the *standardised* steering of `awetrim.identification.controls`: $1.4 \text{ m} \times u_s$ is the steering-tape half-difference (2019 KCU: $u_s = -\text{kcu}/200$, full deflection ± 0.175).
- α is the angle of attack of the wing's centre-panel chord.
- v_a is the apparent wind speed.

- All coefficients are referenced to $S = 19.75 \text{ m}^2$, the EKF/flight area also used by the semi-empirical ROM; the ROM file states it (`aerodynamics.reference_area`).

2.2 Lift and drag

Both coefficients are linear combinations of plain monomials,

$$C_L = c_0 + \sum_i c_i m_i(\alpha, u_p, u_s, v_a, \sigma), \quad C_D = d_0 + \sum_j d_j m_j(\alpha, u_p, u_s, v_a, \sigma), \quad (1)$$

where $\sigma(\alpha)$ is a smooth attached-to-separated switch,

$$\sigma(\alpha) = \frac{1}{2} \left[1 + \tanh\left(\frac{\alpha - \alpha_s}{w}\right) \right], \quad (2)$$

with stall angle α_s and width w (ROM parameters `angle_of_attack_stall`, `width_stall`). Terms multiplied by σ act only past the stall, so $C = C_{\text{att}} + \sigma(C_{\text{sep}} - C_{\text{att}})$ stays linear in its coefficients. Every term is smooth: the ROM evaluates monomials as written (a term may ask for $|m|$ with `abs: true`; the identification never does).

2.3 Bridle pitch and steering

$$\alpha_w = \alpha_b - \theta_b, \quad \theta_b = \theta_0 + \theta_1 u_p, \quad \phi_{a,w} = k_{\phi,s} u_s, \quad (3)$$

with θ_0 , θ_1 , $k_{\phi,s}$ the ROM parameters `angle_pitch_tether_0`, `slope_angle_pitch_tether_depover`, `gain_roll_steering`.

2.4 KCU drag, a separate force

C_D is the *wing's* drag, bridle lines included. The kite control unit is a separate bluff body hanging on the tether at the bridle point (ROM flag `kcu_drag_in_coefficients: false`), with the finite-cylinder law of `awetrim.aerodynamics.kcu_drag`,

$$\mathbf{F}_{\text{KCU}} = \frac{1}{2} \rho (A_{\parallel} |\mathbf{v}_{\parallel}| \mathbf{v}_{\parallel} + A_{\perp} |\mathbf{v}_{\perp}| \mathbf{v}_{\perp}), \quad (4)$$

where $\mathbf{v}_{\parallel}$, $\mathbf{v}_{\perp}$ split the apparent wind about the tether, and the drag areas come from the system file's KCU length, diameter and onboard turbine. The KCU drag enters the total aerodynamic force *and* the bridle resultant $\mathbf{F}_b$ (tether plus the KCU's weight, inertia and drag), so swapping the KCU changes $\mathbf{F}_b$ without re-identifying the wing.

3 Data

Coupled trims (anchors). 341 converged coupled VSM + Billow-wireframe trims (VSM main at 8825a12, artificial viscosity on, 45 panels) from the `_billow9` continuation chains of the 2019 hardware: power-tape sweeps at zero steering for $v_a = 13\text{--}25 \text{ m s}^{-1}$, and steering sweeps at the reel-out and reel-in depower.

Frozen-shape polars. A trimmed chain ties α to u_p almost one to one (α moves by about 1° across the whole v_a range), so the trims alone cannot separate the polar slope from the depower effect. Around every anchor the deformed snapshot is frozen, re-trimmed, and the inflow angle of attack is stepped in the centre-panel body axes from -6° to 22° about the trim (`run_center_sweep_alpha_polar.py`), recording the wing force at each step.

Lens check. A polar is used only if its frozen-shape re-trim reproduces the coupled trim: C_L at the coupled α within 1 %. Steered snapshots with a steering tape of 0.125 m or more re-trim to a different roll/rate state (α up to 2° off, C_L 10–23 % high); their polars are dropped and their coupled trims enter as single samples instead.

Sample. Within the depower band and up to $\alpha = 30^\circ$: 4639 polar samples and 243 coupled-trim samples from 243 anchors.

4 Identification method

4.1 Bridle pitch

With gravity off, the coupled equilibrium is $\mathbf{F}_{\text{wing}} + \mathbf{F}_{\text{KCU}} + \mathbf{F}_t + \mathbf{F}_{\text{in}} = 0$, and the ROM’s bridle resultant is $\mathbf{F}_b = -(\mathbf{F}_{\text{wing}} + \frac{m_w}{m}\mathbf{F}_{\text{in}})$. α_b is its paper Eq. D11 angle about the apparent wind and α_w the centre-panel angle; $\theta_b = \alpha_b - \alpha_w$ is fitted linearly in u_p by ordinary least squares on the 99 unsteered trims in the band.

4.2 Steering gain

$\phi_{a,w}$ is the roll of the wing lift about $\mathbf{v}_a$ relative to $-\mathbf{F}_b$ (paper Eqs. D6–D8, D13); $k_{\phi,s}$ is the least-squares slope through the origin over the 159 steered trims.

4.3 Stall location

α_s and w are chosen first, on a grid ($\alpha_s = 15.5\text{--}18^\circ$, $w = 0.5\text{--}1.5^\circ$), by the cross-validation error of the full C_L library with fixed terms. This separates locating the stall from choosing the terms.

4.4 Candidate terms

Group	Terms	Reason
attached polar	α, α^2	lift slope and curvature
depower	$u_p, u_p^2, \alpha u_p, \alpha^2 u_p$	shape change with the power tape
steering	$u_s^2, \alpha u_s^2, u_p u_s^2$	symmetric in left/right steering
load	$v_a, \alpha v_a, u_p v_a$	deformation grows with dynamic pressure
stall	$\sigma, \sigma\alpha, \sigma\alpha^2, \sigma u_p, \sigma u_s^2$	active past the stall only

There is no α^3 : the ROM’s solvers do not bound α , a cubic C_D went negative beyond the data, and a cubic C_L cannot follow the sharp stall.

4.5 Least squares and selection

For a term set $\mathcal{T}$, the coefficients minimise the sum of squared residuals, $\hat{\mathbf{c}} = \arg \min_{\mathbf{c}} \|\mathbf{y} - X_{\mathcal{T}}\mathbf{c}\|_2^2$, solved with each column scaled to unit RMS for conditioning and mapped back.

Polar samples of one frozen shape are strongly correlated, so the selection score is the *grouped* K -fold cross-validation error with whole anchors held out ($K = 5$):

$$\text{CV}(\mathcal{T}) = \frac{1}{K} \sum_{k=1}^K \sqrt{\frac{1}{|G_k|} \sum_{i \in G_k} (y_i - \hat{y}_i^{(-k)})^2}, \quad (5)$$

where $\hat{y}^{(-k)}$ is fitted without fold G_k . A random split would score interpolation inside a polar rather than prediction at an unseen state. BIC is not used: it treats the correlated samples as independent evidence and keeps accepting terms worth a fraction of a percent.

1. **Forward:** from the intercept, add the candidate that lowers CV most; stop when the best improves it by less than 1 %.
2. **Backward:** drop any term whose removal costs less than the same 1 %; a term picked early can become redundant later.
3. **Extrapolation check:** C_D must stay positive over $\pm 30^\circ$ across the fitted u_p , u_s , v_a ranges, or the identification fails.

4.6 Metrics and uncertainty

Reported are the in-sample RMSE and $R^2 = 1 - \sum r_i^2 / \sum (y_i - \bar{y})^2$ per α band, and the grouped CV RMSE.

The coefficient intervals are an *anchor-cluster bootstrap*: 500 resamples draw anchors with replacement, refit the selected terms on their samples, and report the 2.5/97.5 % percentiles. Resampling whole anchors respects the within-shape correlation that makes ordinary least-squares standard errors far too optimistic. The intervals are *conditional on the selected term set*: the selection is not repeated per resample. θ_b and $k_{\phi,s}$ have one sample per trim, and their standard errors are the ordinary least-squares ones.

5 Results

5.1 Identified coefficients

Table 1: Selected C_L and C_D terms (α in rad, u_p in m, v_a in m s^{-1}) with anchor-cluster bootstrap 95 % intervals.

Coefficient	Term	Estimate	95 % interval	
C_L	1	0.3838	0.3593	0.4074
	αu_p	-0.625	-0.7445	-0.5141
	$\alpha^2 \sigma$	5.406	4.585	6.309
	α	5.459	5.249	5.68
	α^2	-6.898	-7.391	-6.4
	$\alpha \sigma$	-6.061	-6.791	-5.381
	αu_s^2	-39.74	-44.01	-34.93
	$u_s^2 \sigma$	14.2	12.16	16
	u_p	-0.04595	-0.05984	-0.03291
	v_a	0.0009956	0.0007995	0.001174
	σ	0.9579	0.8246	1.101
	$\alpha^2 u_p$	1.82	1.581	2.057
C_D	1	0.0818	0.08147	0.08216
	α^2	1.277	1.256	1.297
	α	0.1191	0.1109	0.1285
	$\alpha^2 \sigma$	-0.8711	-0.964	-0.7737
	$u_p \sigma$	-0.02216	-0.02959	-0.01426
	αv_a	0.001259	0.000725	0.001685
	$\alpha \sigma$	0.4555	0.3783	0.5291
	u_s^2	-0.5255	-0.6016	-0.4706
	αu_s^2	2.474	1.535	3.861

Table 2: Selection path: the term added (+) or removed (−) at each step and the grouped CV RMSE after it.

C_L			C_D		
Step	Term	CV RMSE	Step	Term	CV RMSE
0	intercept	0.31	0	intercept	0.0977
1	$+\alpha u_p$	0.196	1	$+\alpha^2$	0.0106
2	$+\alpha^2 \sigma$	0.0814	2	$+\alpha$	0.00664
3	$+\alpha$	0.0362	3	$+\sigma$	0.00572
4	$+\alpha^2$	0.0282	4	$+\alpha^2 \sigma$	0.00521
5	$+\alpha \sigma$	0.0243	5	$+\sigma u_p$	0.00499
6	$+\alpha u_s^2$	0.0233	6	$+\alpha v_a$	0.00484
7	$+\sigma u_s^2$	0.0226	7	$+\alpha \sigma$	0.00472
8	$+\sigma u_p$	0.0216	8	$+u_s^2$	0.00465
9	$+u_p$	0.0212	9	$+\alpha u_s^2$	0.00459
10	$+v_a$	0.0207	10	$-\sigma$	0.00462
11	$+\sigma$	0.0205			
12	$+\alpha^2 u_p$	0.0202			
13	$-\sigma u_p$	0.0202			

Table 3: Fit quality per angle-of-attack band (all fitted samples).

Samples	n	RMSE C_L	RMSE C_D	R^2 (C_L / C_D)
all	4882	0.0201	0.00458	0.9958 / 0.9978
$\alpha < 13^\circ$ (operating)	2915	0.0125	0.00307	0.9983 / 0.9906
$13^\circ \leq \alpha < 19^\circ$ (stall)	960	0.0381	0.00759	0.8212 / 0.9457
$\alpha \geq 19^\circ$ (post-stall)	1007	0.0112	0.00441	0.7349 / 0.9818
Grouped 5-fold CV RMSE		0.0202	0.00462	

Stall. $\alpha_s = 17^\circ$, $w = 1.5^\circ$. Every coupled polar peaks near $C_L \approx 1.2$ at $15.5\text{--}16.7^\circ$ and drops within about 2° to a plateau near 0.87. The stall band carries the largest residual, because one polynomial family has to follow a sharp drop; the operating range below 13° is fitted to a C_L RMSE of about 0.012.

Bridle pitch. $\theta_0 = -0.345 \pm 0.0046 \text{ rad}$ and $\theta_1 = 0.258 \pm 0.0024 \text{ rad m}^{-1}$ (residual 0.24° RMS): $\theta_b = 5.35^\circ$ at 1.7 m and 11.25° at 2.1 m. With the KCU drag inside C_D , the bridle resultant would lie along the tether and θ_b would be about 2.5° larger (Fig. 2). The semi-empirical ROM, calibrated in that convention, has a 45 % steeper slope.

Steering gain. $k_{\phi,s} = -0.530 \pm 0.0043 \text{ rad per unit } u_s$ (residual 0.32° RMS), against -0.689 for the semi-empirical ROM in the same units.

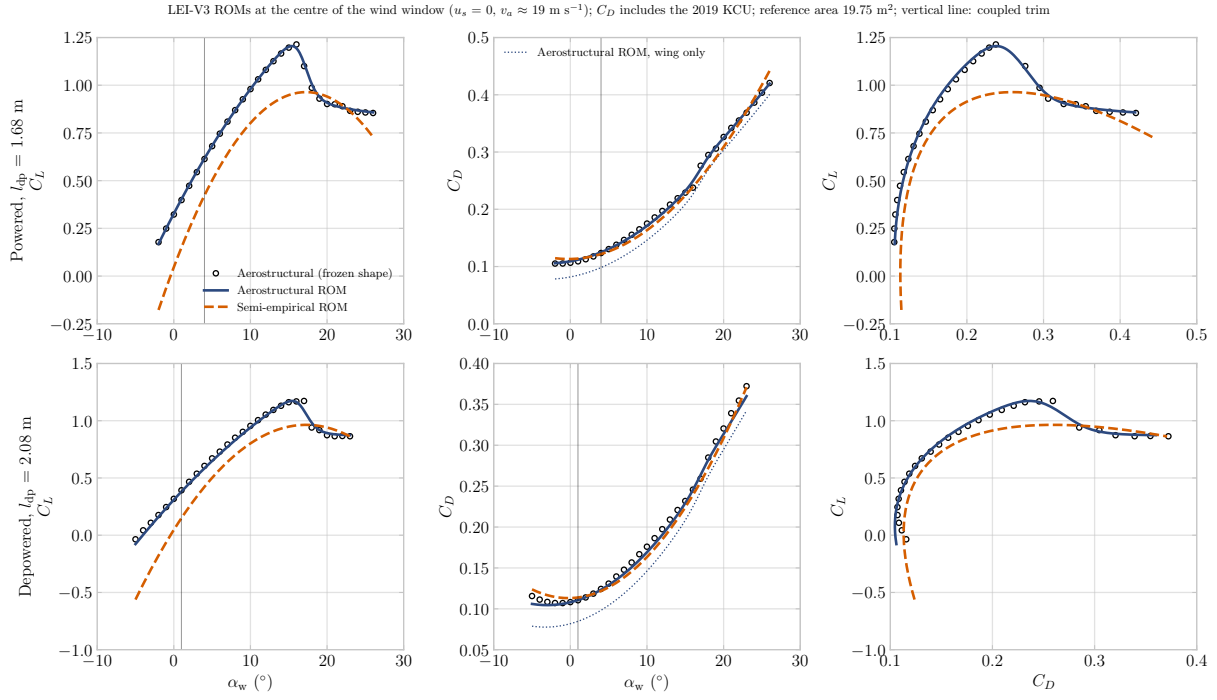


Figure 1: Lift and drag polars at the powered and depowered setting ($u_s = 0$, $v_a \approx 19 \text{ m s}^{-1}$, C_D with the 2019 KCU): coupled frozen-shape data, the aerostructural ROM and the semi-empirical ROM.

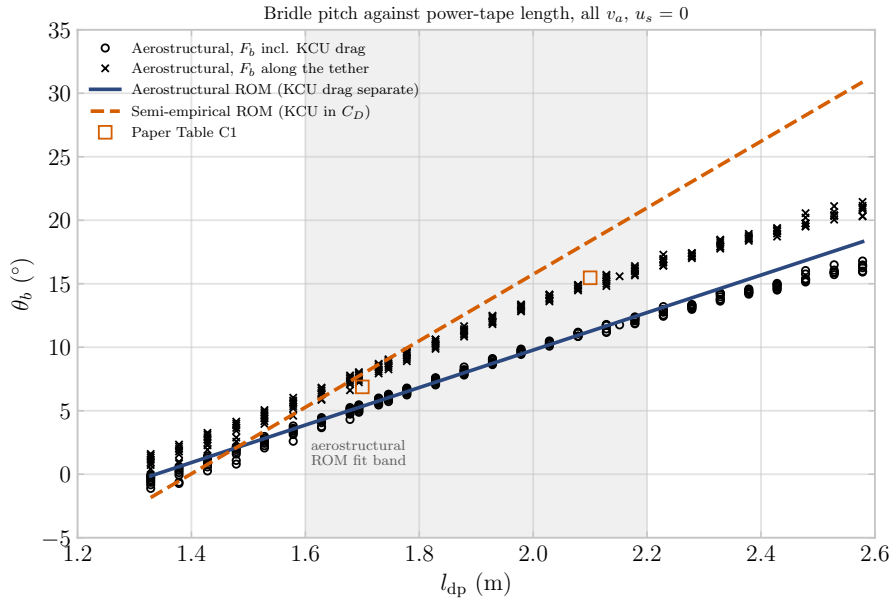


Figure 2: Bridle pitch against power-tape length: coupled trims in both bridle conventions, the aerostructural relation and the semi-empirical one.

5.2 Check against the coupled trims

Each coupled trim is re-solved with the ROM's own quasi-steady solver at the same condition: centre of the window, same wind, u_p , u_s , gravity off, tetherless (`validate_rom.aerostructural.py`).

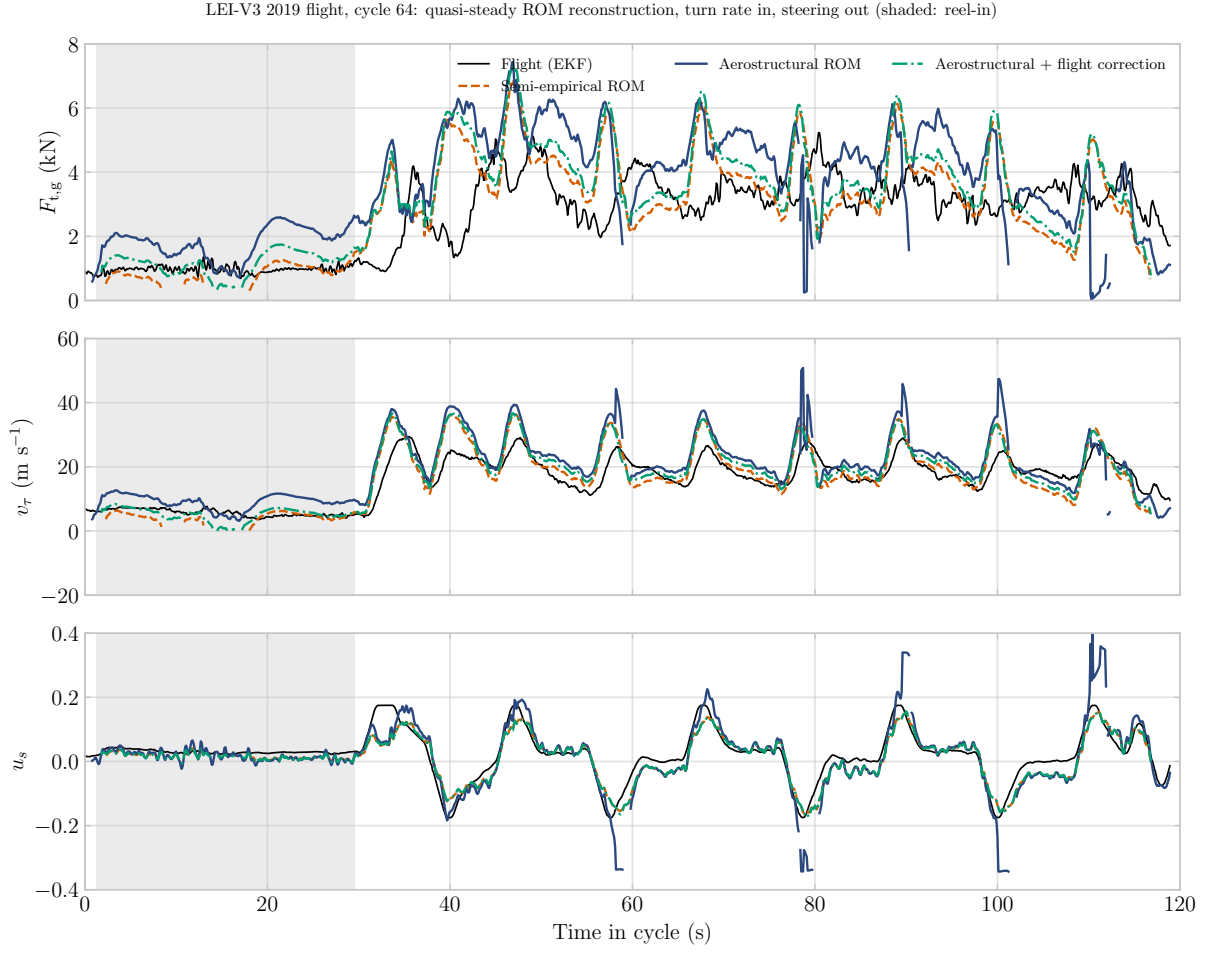


Figure 3: Cycle 64 (held out) with the measured turn rate as input and the steering solved: ground tension, tangential speed and the steering each ROM needs, against the EKF reconstruction (solid black, logged u_s with the rig's zero offset). Shaded: reel-in. Gaps are samples without a converged root.

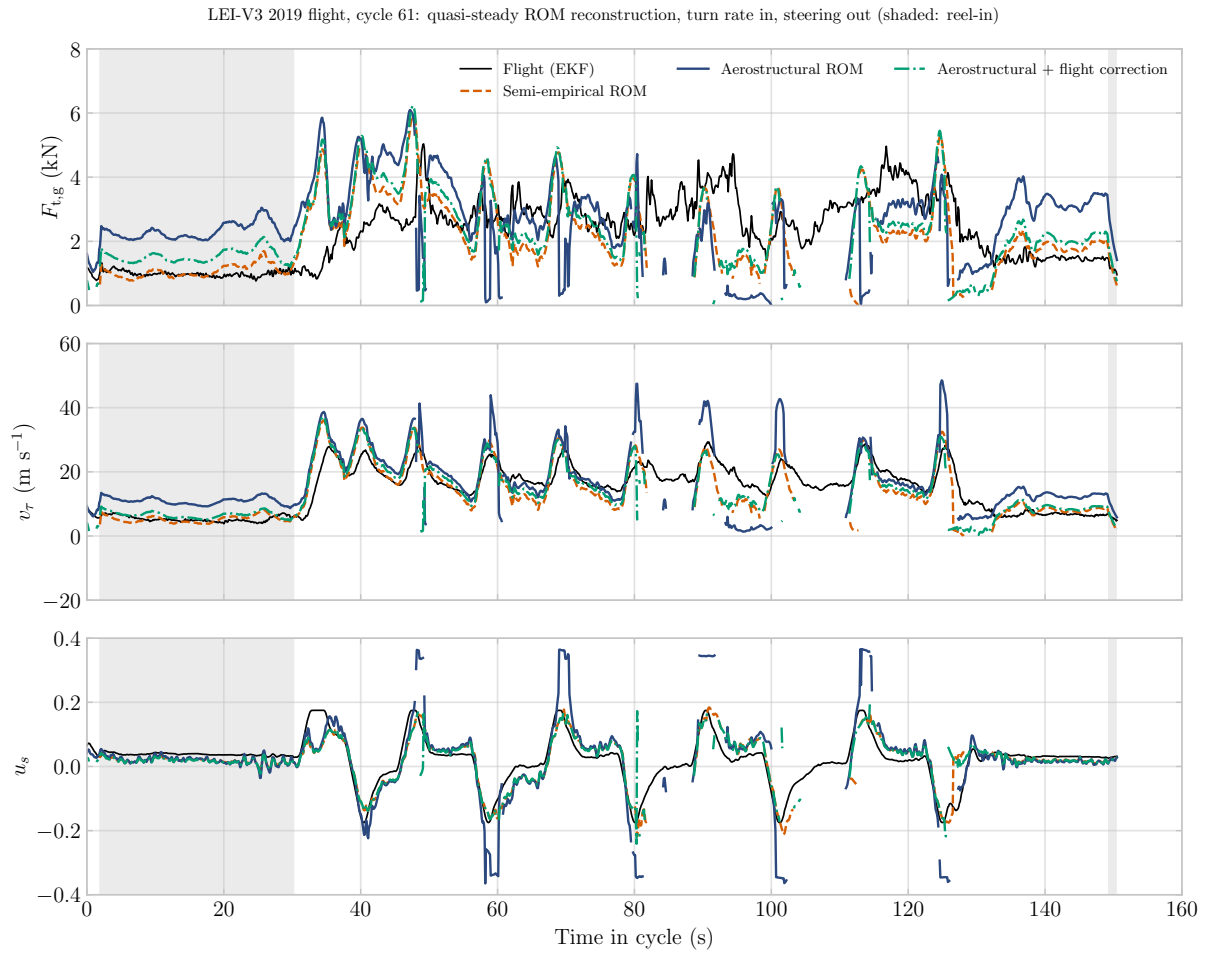


Figure 4: As Fig. 3 for cycle 61.

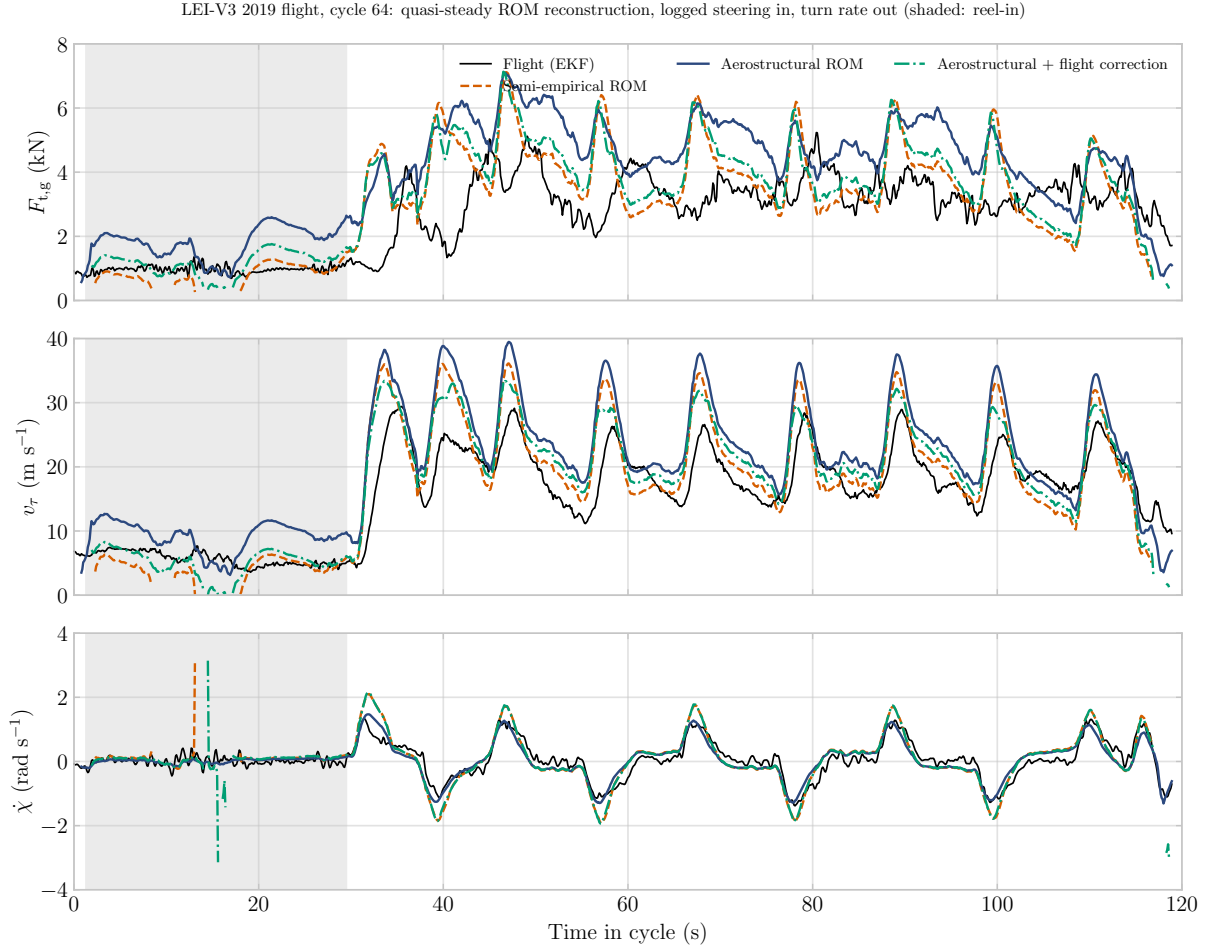


Figure 5: Cycle 64 with the logged steering as input.

With the measured turn rate as input, the simulation-only ROM reproduces the held-out reel-out to +7% in tension and +10% in speed, with no flight data in it at all; with the logged steering as input, to +23% and +14%. It still flies somewhat faster than the kite did, which is the direction of the VSM’s known drag deficit [3], but the deficit is a fraction of what the uncorrected frame had made it look like. Its solved steering has slope 1.6 against the logged steering: the aerostructural roll gain (−0.53) is too weak, by exactly that factor (Sect. 7). The semi-empirical rows are the paper’s ROM *recalibrated* on the corrected frame with the step-3 method of Sect. 7 (θ_b , $C_{D,0}$, the $|u_s|$ drag and the roll gain; `refine_rom_flight_output_error.py --rom semi_empirical`, the paper’s values kept in the file’s comments): reel-out −10%/−6% with the turn rate in and −6%/−1% with the logged steering in, its reel-in tension within −6% but its reel-in speed −18%. Its recalibrated gain, −0.69, is the paper’s. After this comparison the semi-empirical ROM was removed from the repository (archived in git history; `config_paths` keeps its path): the flight-corrected aerostructural ROM is the better-founded model and is now the default in every LEI-V3 system file. The rows and figures here are kept as the record.

Reel-in is where the aerostructural polar fails: that ROM doubles the tension, and the flight-corrected one (aerostructural-flight rows, Sect. 7), which brings the reel-out within a few percent in both modes, still reads +33% there. The recalibrated semi-empirical ROM does better on reel-in tension but not on reel-in speed; Sect. 9 discusses why reel-in resists.

The tether model is not what separates the ROMs. With the Williams tether the reel-out errors move by a few points (aerostructural +14%/+12%, flight-corrected +6%/+4%, semi-empirical −20%/−11%) and the steering regressions are unchanged. On reel-in it helps the flight-corrected ROM (+33% → +21% tension) and costs the semi-empirical one most of its roots

(34 % solved).

6 The bridle-angle frame

The ROM takes two angles of the bridle resultant $\mathbf{F}_b$ about the apparent wind: the bridle pitch α_b (paper Eq. D11), from which $\alpha_w = \alpha_b - \theta_b$, and the bridle roll ϕ_b (Eqs. D6–D8), the reference the steering roll $k_{\phi,s}u_s$ is measured from. In `awetrim.system.kite` both were computed by rotating $\mathbf{F}_b$ into an Euler-angle “aerodynamic” frame built from the pitch and yaw of $\mathbf{v}_a$. Checked numerically on a flown state with sideslip (the apparent wind having a component normal to the course), that frame’s x axis was not along $\mathbf{v}_a$: $\hat{\mathbf{x}}_A \cdot \hat{\mathbf{v}}_a = -0.97$, and flipping the sign of the yaw angle gives -1.0000 exactly. The yaw sign was mirrored for the rotation order used, so the frame sat at twice the sideslip angle off $\mathbf{v}_a$ in the tangent plane, with the normal axis off by the same amount. The lift direction, built directly from the $\mathbf{v}_a$ vector elsewhere in the same class, was right.

The error is zero at zero sideslip: the centre-window trims the aerostructural ROM was identified on and checked against (Sect. 5) have none. In crosswind flight the apparent wind has a normal component of $3\text{--}5\text{ m s}^{-1}$ at 20 m s^{-1} , a sideslip of $7\text{--}14^\circ$, and the frame error let the normal component of $\mathbf{F}_b$ — the KCU’s inertia in a turn, some 600 N on 5 kN — leak into the pitch and the roll. On cycle 64 the bridle pitch came out $2.3\text{--}3.0^\circ$ too high at samples of one sideslip sign and correspondingly low at the other, and the bridle roll about 1° off. Since the lift direction was consistent, the wing’s roll relative to the *true* bridle roll was no longer $k_{\phi,s}u_s$ but $k_{\phi,s}u_s$ plus a sideslip-dependent offset, which is what made the solved steering noisy and bimodal: before the fix, in the $|u_s| > 0.12$ bin, the aerostructural ROM solved 0.34 for a flown 0.16, the semi-empirical one 0.16; after it, both solve the flown value and the regression of solved on logged steering has correlation 0.92–0.93 for every ROM.

The fix builds the paper’s frame from the $\mathbf{v}_a$ vector (`Kite.force.in.wind.frame`: $x = -\hat{\mathbf{v}}_a$, $y = -\mathbf{e}_n$, $z = \mathbf{e}'_r$), as the lift direction already did, and a regression test now holds the bridle pitch to Eq. D11 and the bridle roll to the roll of the resultant on a KCU-loaded case with sideslip. What is affected: every flight validation and every flight calibration done before the fix, including the paper’s semi-empirical ROM, whose output-error calibration absorbed the mean of the error (its reel-out tension went from $+11\%$ on the old frame to -19% on the correct one without changing a parameter; it has been recalibrated, Sect. 5); and the optimisations, whose stall margin acts on an α that was up to 3° off in the lobes (rerun, Sect. 8). The identification of the aerostructural ROM is not affected.

7 Flight correction

The third LEI-V3 ROM, `rom_config_aerostructural_flight_corrected.yaml` (`--rom aerostructural_flight`), starts from the aerostructural one and refits six parameters to the 2019 flight; everything else (lift polynomial, stall, reference area, separate KCU drag) stays simulated:

$$\theta_b = \theta_0 + \theta_1 u_p, \quad (6)$$

$$\phi_{a,w} = k_{\phi,s} u_s, \quad (7)$$

$$C_D = C_{D,AS}(\alpha, u_p, u_s, v_a) + \Delta C_{D,0} + c_p u_p + c_s u_s^2. \quad (8)$$

The bridle pitch absorbs the as-built versus flown bridle geometry and the calibration from logged depower to power-tape length. The drag correction has single-input terms only, no products; steering enters squared because the drag cannot depend on the turn direction, and $|u_s|$ would put a kink at zero. The roll gain and the steering drag are the two things steering does to the wing, so they are identified together, from the same flown states. Cycles 60–67, the flight-validation cycles, are held out of every step.

7.1 Steps 1–2: equation error on the EKF coefficients

Each flown sample gives the EKF lift coefficient, the total aerodynamic drag of wing, bridle and KCU (minus the ROM’s own KCU drag model, so the target is in the ROM’s C_D convention), and the bridle angle of attack α_b of paper Eq. D11, computed on the bridle resultant the ROM builds from the EKF’s bridle-element force. That is a force angle, not a vane reading.

1. **Bridle pitch from lift.** The angle α^* at which the aerostructural $C_L(\alpha^*, u_p, u_s, v_a)$ equals the flight C_L is found per sample; $\alpha_b - \alpha^*$ is the θ_b the flight needs. A correction $\Delta\theta_0 [+ \Delta\theta_1 u_p]$ to the aerostructural relation is fitted to it, with the slope kept only if the cross-validation (folds by cycle) asks for it. It did: the flight needs a steeper depower authority than the simulations (Table 6, top-left of Fig. 6).
2. **Drag at equal lift.** $C_{D,AS}$ is evaluated at the same α^* : the correction is to the drag *polar* $C_D(C_L)$. Evaluating it at $\alpha_b - \theta_b$ instead would be circular: α_b is essentially the flight glide angle $\arctan(C_D/C_L)$ plus wing-mass terms, so the flight drag would pick the angle it is compared at.
3. **Roll gain.** Paper Eq. 43, $\phi_{a,w} = k_{\phi,s} u_s$: the roll of the wing force about $\mathbf{v}_a$ relative to the bridle resultant (Eqs. D6–D8, as on the coupled trims), the wing force being minus the bridle resultant plus the wing’s inertia and weight. Fitted through the origin against u_s delayed by 0.3s, the EKF’s own steering-input lag ($R^2 = 0.91$; without the delay $k_{\phi,s} = -0.751$). A free intercept comes out at -0.06° , an independent check of the 2019 steering zero offset. The flight gain, -0.765 , is about 44 % stronger than the aerostructural -0.53 and close to the semi-empirical -0.69 .

The first two are linear least squares with forward–backward selection on grouped cross-validation RMSE, the third a one-parameter least squares. The intervals are a cycle bootstrap (500 draws) repeating all three.

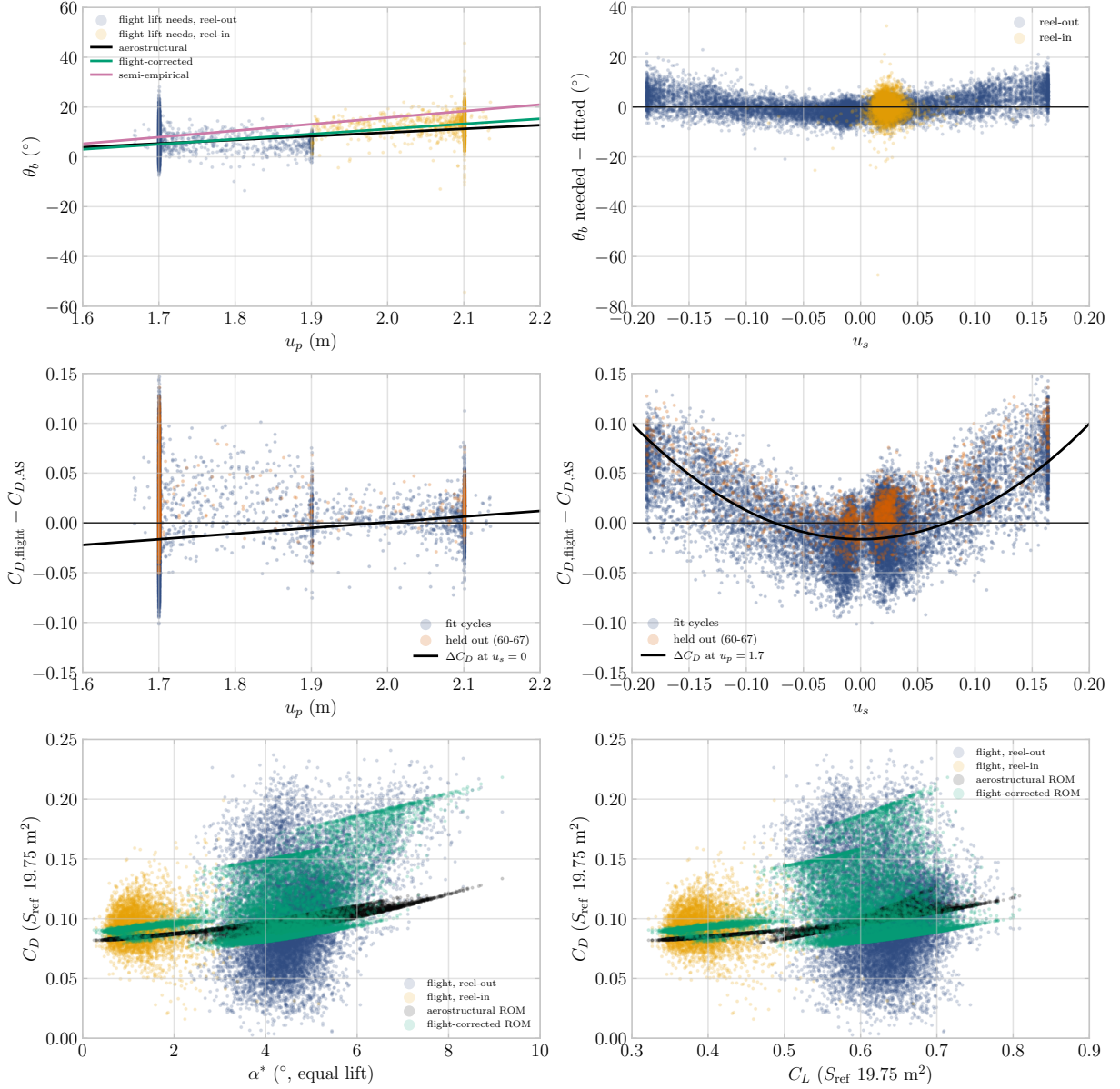


Figure 6: Steps 1–2 on the 2019 flight. Top: the θ_b the flight lift needs against u_p , with the aerostructural, flight-corrected (steps 1–2) and semi-empirical relations, and its residual against steering. Middle: the drag polar residual at equal lift against u_p and u_s , fit and held-out cycles. Bottom: drag against α^* and the drag polar.

Two cautions on what these fits mean. The measured roll is almost entirely the wing’s inertia term: in the EKF’s own force balance $\mathbf{F}_w + \mathbf{F}_b = m_w(\mathbf{a} - \mathbf{g})$, so the roll of the wing force relative to the bridle resultant is $m_w a_n / |\mathbf{F}|$, and it correlates with u_s because steering turns the kite. Without the inertia term the fitted gain collapses to -0.09 ($R^2 = 0.06$), and it scales with the wing mass assumed (11 kg: -0.60 ; 17 kg: -0.91). It is a turn-response-per-steering times a mass split, not an aerodynamic measurement on its own, and it is the roll the wing *had* at the turn it made, against every other normal force the point-mass model puts on the kite; Sect. 7.3 shows it reads 20 % too strong once compared with the turn-rate law. It serves as the starting value and as a check of the sign and order of magnitude. And the equal-lift drag rise with steering sits where the EKF is least reliable: at $|u_s| > 0.12$ its innovation statistics are 2–3 $\times$ worse than in straight flight and the inertial force is of the order of the drag.

7.2 Step 3: output error on the quasi-steady solve

Steps 1–2 match the flight coefficients at the flown states, but the quasi-steady reconstruction with them still over-predicts the held-out reel-out (+19 % tension, +12 % speed). The flown samples are not equilibria: $|m\dot{v}_\tau|/D$ is 0.1 at the median and 0.3 at the 90th percentile on reel-out, and the EKF coefficients are instantaneous, whereas the ROM is used as a quasi-steady model. Step 3 therefore refits all six parameters together on the output the ROM is used for: the quasi-steady ground tension and tangential speed with the measured turn rate in and the steering solved, on the rigid tether of the optimisations (`refine_rom_flight_output_error.py`). Tension and speed barely see the roll gain, which mostly sets how much steering the solve uses, so the gain gets its own condition, the turn-rate law the paper defines the steering by: the solved steering must regress 1:1 on the logged steering (zero offset, 0.3 s lag) over the reel-out training samples, one residual 5 (slope -1) with its exact Jacobian from the per-sample sensitivities. The condition pins the gain only if drag cannot buy steering: left free, the fit met the slope with the gain at -0.77 and the total u_s^2 drag driven to -0.25 (a negative steering drag, which makes the solve fly faster and roll less at high steering), so the steering-drag coefficients are bounded at zero in total, C_D non-decreasing with steering. Holding the gain with the measured aerodynamic roll of step 2 instead was tried first and gives -0.75 ; Sect. 7.3 shows why that is 20 % too strong. 1200 samples from 24 training cycles enter, half reel-out, half reel-in. Each sample’s root is found by Newton’s method, warm-started from its last root, and its derivative with respect to the parameters follows from the implicit-function theorem, so the outer Gauss–Newton (trust region) has exact Jacobians. A sample that loses its root, or whose root lies past the stall, is charged a fixed error instead. One large NLP with every sample’s balance as a constraint was tried first and is infeasible: reel-in samples sit at the fold of the radial balance (Sect. 9) and lose their root when the parameters move. A variant that let a steering error (solved against logged u_s) hold the gain instead of the measured roll drove it to a non-physical -4.9 , the logged steering explaining the turn poorly (lag; gravity and tether turn the kite at zero steering), and is not used.

Table 6: Bridle pitch, drag correction and roll gain. Drag terms are corrections to the aerstructural C_D (whose own u_s^2 coefficient is about -0.53). Steps 1–2 with the cycle-bootstrap 95 % interval; step 3 is the adopted ROM.

Parameter	Aerostructural	Steps 1–2 (95 % CI)	Step 3	Step 3, reel-out only
θ_0 (rad)	−0.345	−0.518 [−0.543, −0.491]	−0.587	−0.512
θ_1 (rad m ^{−1})	0.258	0.357 [+0.343, +0.370]	0.41	0.368
$\Delta C_{D,0}$	0	−0.113 [−0.130, −0.095]	−0.182	−0.263
$\Delta C_D: u_p$ (m ^{−1})	0	0.0569 [+0.049, +0.065]	0.103	0.166
$\Delta C_D: u_s^2$	0	2.9 [+2.868, +2.929]	2.82	0.525
$k_{\phi,s}$ (rad)	−0.53	−0.765 [−0.772, −0.758]	−0.666	−0.701

Table 7: Quasi-steady reconstruction of the 24 training cycles (median errors, steering solved, rigid tether) with the step 1–2 and the step 3 values; u_s rms against the logged steering plus the zero offset.

Phase	Parameters	F_t error (%)	v_τ error (%)	u_s rms	Solved (%)
reel-out	steps 1–2	10.4	4.81	0.036	100
	step 3	−5.98	−4.36	0.039	87
	step 3, reel-out only	−2.3	−6.08	0.038	92
reel-in	steps 1–2	23.3	6.66	0.012	100
	step 3	17.1	−13.2	0.01	99

7.3 Result

The steering gain against the turn-rate law. The empirical turn-rate law $\dot{\chi} = K v_a u_s + c_g g \cos \beta \sin \chi / v_\tau$ fitted to the 2019 reel-out (logged u_s with the zero offset and the 0.3 s lag) gives $K = 0.343 \text{ rad m}^{-1}$ and $c_g = 0.47$; a point mass has $c_g = 1$, so the kite turns under gravity about half as much as its mass would, its yaw stiffness resisting course changes it was not steered into. The same regression on each ROM’s turn rate in the logged-steering mode (Table 8) is the check of the gain: with the gain from the measured roll (-0.75) the flight-corrected ROM gave $K = 0.41$, 20 % above the flight, with the aerostructural gain (-0.53) $K = 0.27$; and every ROM has $c_g \approx 1.1\text{--}1.2$, i.e. $2\text{--}2.5\times$ the flight’s gravity-driven turn. The two are coupled: the measured roll is the roll needed to make the measured turn *against* that exaggerated gravity turn, which is why it over-reads the gain. The turn-rate-law condition of step 3 sets the gain where the solved steering regresses 1:1 on the logged one: -0.67 for the flight-corrected ROM and -0.69 for the recalibrated semi-empirical one, whose held-out K are 0.377 and 0.383. The gravity-turn excess is a property of the point-mass force model (Sect. 9).

Table 8: Turn-rate law fitted to the flight and to each ROM’s reconstructed turn rate (logged steering in, reel-out, cycles 60–67).

$\dot{\chi}$ of	K (rad m ^{−1})	c_g	R^2
flight	0.343	0.47	0.875
semi-empirical	0.383	1.22	0.889
aerostructural	0.270	1.00	0.878
aerostructural-flight	0.377	1.17	0.875

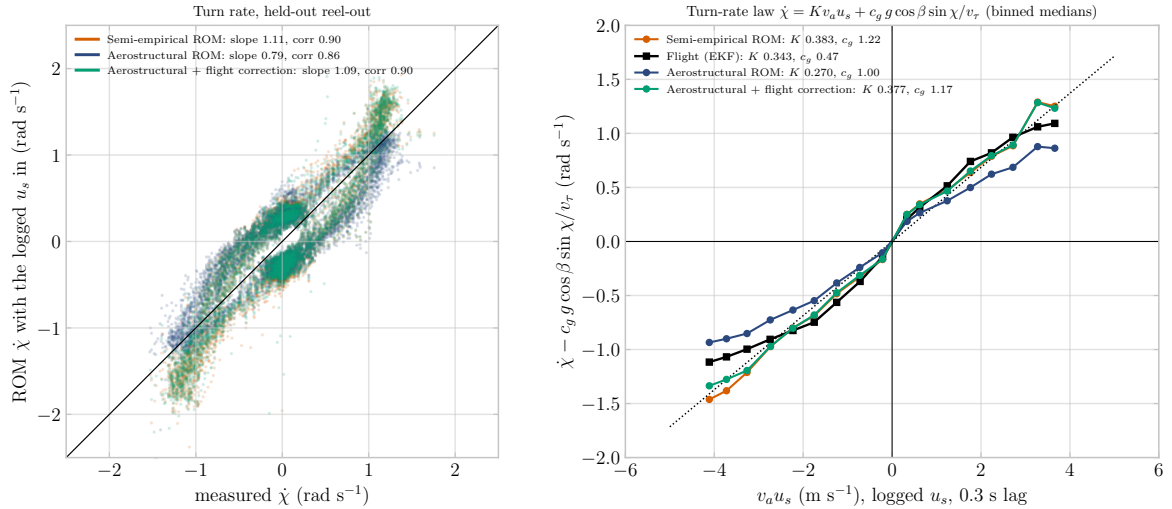


Figure 7: The turn rate of each ROM with the logged steering as input against the measured one (left, held-out reel-out samples), and the turn-rate law with the gravity term removed, binned in $v_a u_s$ (right): the flight against the ROMs. The tuned ROMs sit on the flight’s K within 10 %; the aerostructural gain under-turns by a third. The flight’s response bends below linear beyond $|v_a u_s| \approx 3$, a saturation at full steering that a linear gain cannot follow, which is where the tuned ROMs’ peak turn rates overshoot.

Held out. On the held-out cycles (Table 5, Figs. 3 and 4), with the turn rate in, the corrected ROM reproduces the reel-out tension and speed to -3% and -1% (median) and the reel-in to $+32\%$ and 0% , with its solved steering on the logged one (slope 0.88, correlation 0.89). With the logged steering in, it reproduces the reel-out to $+1\%$ and $+4\%$ (semi-empirical: -6% and -1%) and the turn rate with correlation 0.90, and the reel-in tension again to $+33\%$; with the Williams tether the reel-in tension error drops to $+20\%$. 93 % of the training samples keep a valid root

at the step-3 parameters. Refitting on the reel-out samples alone (last column of Table 6) moves θ_b by 0.2° at the powered and 0.7° at the depowered setting and the gain by 5 %: the reel-out parameters do not depend on the reel-in samples' pull.

Three cautions on the parameters themselves:

- **$\Delta C_{D,0}$ and c_p are nearly interchangeable.** The flown u_p takes essentially two values (1.7 m and 2.1 m), so only their sum at those two settings is identified; the split between them is not.
- **Lift terms without α would not be identifiable either.** A lift offset $\Delta C_{L,0}$ or $c_{L,p}u_p$ raises the lift at the operating point exactly as a pitch shift $\Delta\theta_0$ or $\Delta\theta_1u_p$ does; tension and speed fix the lift and drag at the operating point, not the α they occur at, and the only independent α measurement, the EKF's bridle vane, sits 4° from the force-based angle. Where the depower correction lives (pitch or polar) is a modelling choice; it matters only where α itself matters, the stall margin.
- **Steering drag is not identified.** At equal lift the flight drag rises with steering by $c_s \approx +2.9$; the output error on both phases keeps +2.3 of it, on the reel-out samples alone it goes to the bound at zero with the zero-steering drag level rising by 0.03 to compensate. Tension and speed cannot tell a steering drag from a drag level, so c_s is a range, 0 to about 2.3, tied to the drag level, not a value; a u_s^2 lift loss, the natural companion term, would sit in the same unidentified regime.

8 Optimisations

The three ROMs were run through the reel-out downloop optimisation (`downloop_pattern.py`: one figure-eight at 8 ms^{-1} and 100 m, logarithmic wind, rigid tether, power-tape length bounded to 1.6–2.2 m, a 3° stall margin on α), on the optimisation hardware of `system.yaml` and on the flown 2019 hardware (KCU 22 kg with its turbine), all on the corrected frame. This is an application of the tuned ROMs, not part of their identification or validation: no optimisation result enters any fitted parameter, and none of these loops was flown, so nothing here can be checked against data. The optimiser lets each ROM pick its own loop (Fig. 8), so the comparison is of what each ROM thinks the kite can do, not of one path flown with three polars.

Table 9: Optimised downloop reel-out per ROM: mean power, mean tension, mean tangential speed and loop time.

System	ROM	$\bar{P}$ (kW)	$\bar{F}_t$ (kN)	$\bar{v}_\tau$ (ms^{-1})	T (s)
optimisation hardware (KCU 8.4 kg)	semi-empirical	7.15	4.98	23.9	9.4
	aerostructural	10.42	6.18	27.1	12.8
	aerostructural-flight	10.78	6.30	27.7	12.3
flown 2019 hardware (KCU 22 kg + turbine)	semi-empirical	5.05	4.35	22.1	12.8
	aerostructural	5.49	4.44	23.4	19.3
	aerostructural-flight	5.90	4.62	22.9	15.5

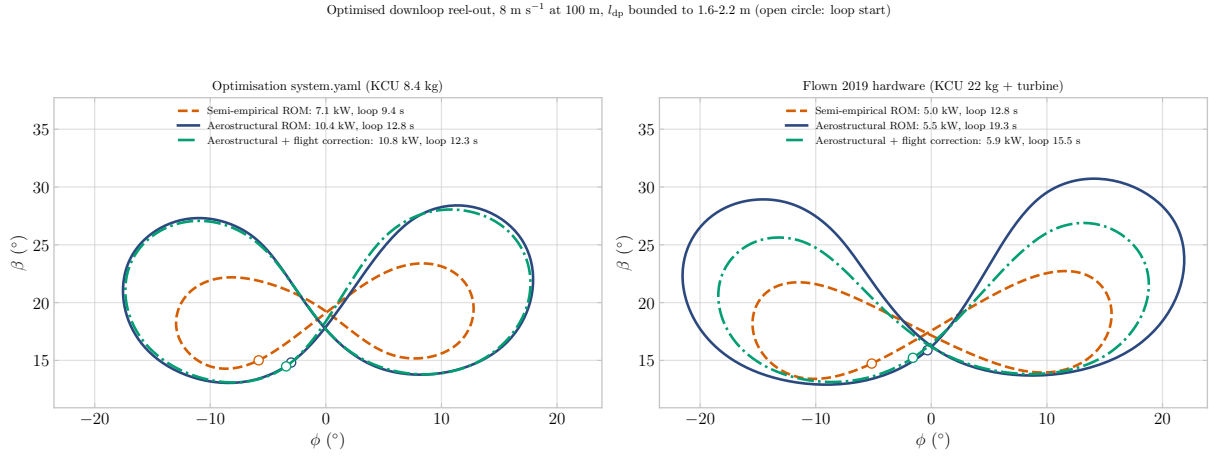


Figure 8: The optimised loops in azimuth and elevation, per ROM and hardware.

On the flown 2019 hardware the three ROMs are within 5.0–5.9 kW, the flight-corrected one highest and the semi-empirical lowest, in the order of their reel-out tension in the validation. On the lighter optimisation hardware the two aerostructural ROMs reach 10.4 kW and 10.8 kW against the semi-empirical 7.1 kW, and fly taller loops (elevation up to 28° against 23°). The difference is the polar: the aerostructural one has the higher lift-to-drag ratio (it flew the validation 10% faster) and a steering that costs it little drag, the semi-empirical one carries more drag at the same lift and, after recalibration, a larger $|u_s|$ drag; with the KCU at 8.4 kg the gravity-driven part of the turn shrinks, and the higher- L/D polar lets the optimiser pull the loop up and fly it faster. The spread between the two aerostructural ROMs ($\pm 2\%$) is within what the flight correction can resolve; the spread to the semi-empirical ROM (about a third lower) is a difference of polar shape, not of calibration, and is not resolved by the 2019 flight, which never flew that hardware or that pattern.

9 Limitations and open points

- **Reel-in reads as speed-unstable.** At fixed controls, every flown 2019 reel-in state reconstructed by the semi-empirical ROM has $\partial \dot{v}_\tau / \partial v_\tau > 0$, while reel-out reads negative. A $\dot{v}_\tau(v_\tau)$ sweep at flown states, controls fixed and the radial balance solved for the tension at each speed (Fig. 9, `sweep_vdot_tangential.py`), shows why. On reel-out it reproduces the paper’s Fig. 6: a stable fast root at the reconstructed speed and an unstable post-stall root below it at the fastest state; at the slower reel-out states (12 m s^{-1} to 15 m s^{-1} measured) the branch through the reconstructed root already ends at a fold below the measured speed, the reconstruction sitting low (-7% in the median, Table 5). On reel-in there is no fast root at any state. Each branch *ends*, at $v_\tau \approx 3\text{--}7 \text{ m s}^{-1}$, where no tension can balance the radial forces any more: tension also pitches the bridle resultant, so the radial residual has a maximum over tension that crosses zero at that speed (a fold). The flown reel-in states sit within about 1 m s^{-1} of the fold, where $\partial \dot{v}_\tau / \partial v_\tau > 0$. In this model the kite reels in as fast as the radial balance allows. The straight, rigid tether is weakest exactly at this low tension, so the sweep was repeated with the Williams tether, the radial row solved together with the ground closure (Fig. 10). The fold survives: the reel-in branches still end at $v_\tau \approx 4\text{--}7 \text{ m s}^{-1}$ with $\partial \dot{v}_\tau / \partial v_\tau > 0$. The fold is a property of the kite model, not of the tether model.
- **Steering range.** The steering terms are identified up to $|u_s| \approx 0.18$. Beyond it the lift loss with steering ($\propto \alpha u_s^2$) extrapolates quadratically, and tight seed paths that need more steering are not followable.
- **Stall.** The stall is that of VSM with artificial viscosity on frozen deformed shapes; it locates

the lift maximum but is the least validated part of the model.

- **Centre of the window, gravity off.** Position-dependent aerodynamics (sideslip, gravity-induced attitude) are outside the data.
- **Drag deficit.** Most of the apparent deficit was the frame error of Sect. 6; what remains (+7 % tension, +10 % speed on reel-out) is within what the flight correction absorbs, but it is absorbed by θ_b and a drag level, which the data cannot tell apart from a lift level.
- **Reel-in.** No ROM reproduces both its tension and its speed: the aerostructural polar doubles the tension even after the flight correction (+33 %), the recalibrated semi-empirical ROM gets the tension (−7 %) but not the speed (−18 %). The flown reel-in states sit at the fold above, with the kite reeling in near the speed the radial balance allows; a quasi-steady point-mass model has one knob there, the tension, and the winch transient the kite is actually in is outside it.
- **Flight data.** The EKF reconstruction is a model output too: its forces depend on its tether model, its coefficients are instantaneous at states that are not equilibria, and its innovation statistics are 2–3× worse in turns than in straight flight. The cycle bootstrap intervals quoted here cover cycle-to-cycle variation only, not these systematic effects, and the 3–4° by which the equation-error and output-error θ_b differ is a better measure of what the data pin down.

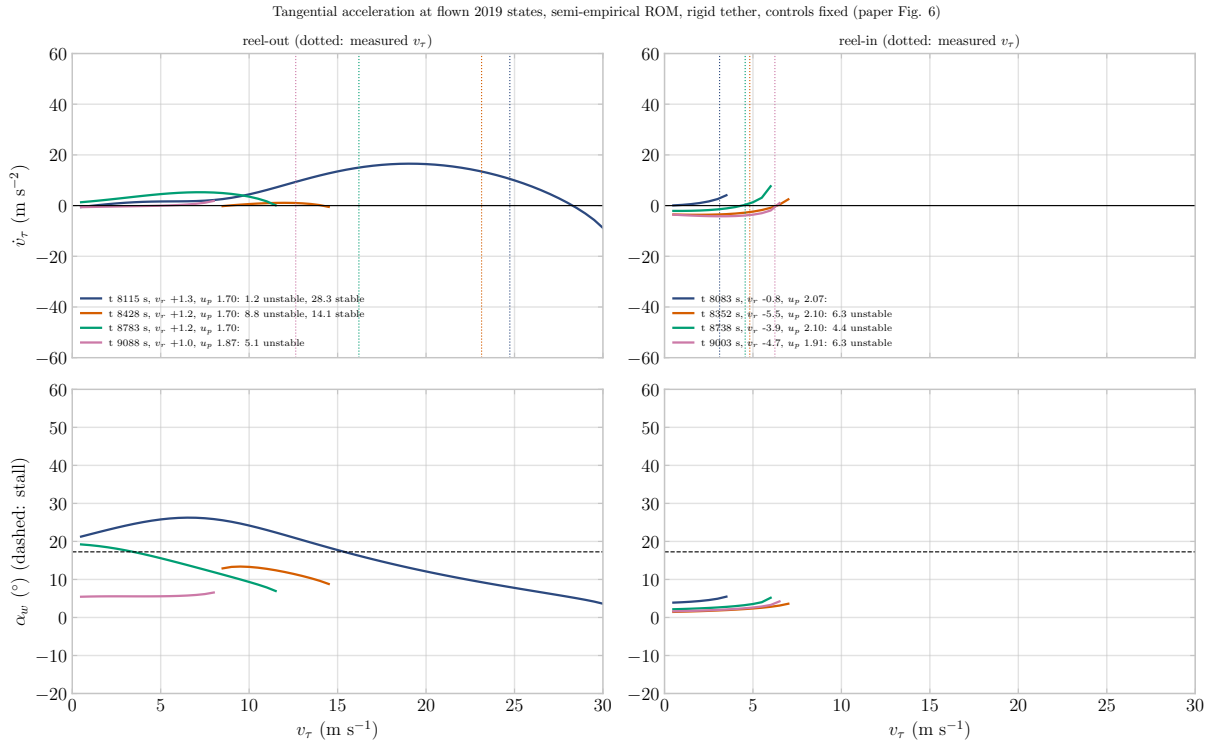


Figure 9: Tangential acceleration and wing angle of attack against tangential speed at flown 2019 states (semi-empirical ROM, controls fixed, tension from the radial balance). Reel-in branches end at the fold of the radial balance.

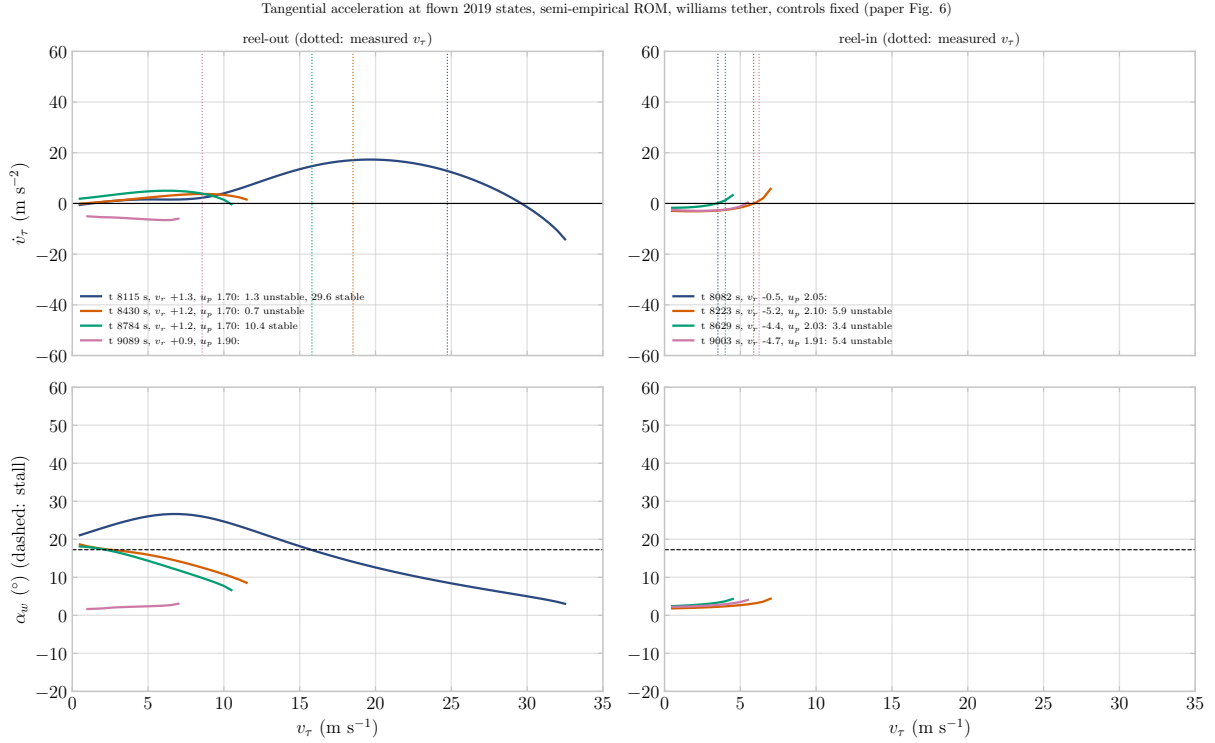


Figure 10: As Fig. 9 with the Williams tether: the radial row and the ground closure are solved at each speed for the kite-end tension, the tether length and the last element's angles. The reel-in fold remains.

10 Reproduction

From the repository root:

1. `scripts/personal/wes-quasi-steady/run_center_sweep_alpha_polar.py` (pinned VSM) for the frozen-shape polars;
2. `scripts/identification/identify_rom_aerostructural.py`, which writes the ROM file and the tidy datasets;
3. `scripts/identification/validate_rom_aerostructural.py` and `plot_rom_comparison.py`;
4. `scripts/reduced-order-model/validation/validate_quasi_steady_state_v3.py` with `--rom<rom>--steeringsolved|measured`, then `plot_rom_flight_validation.py`;
5. for the flight correction: `scripts/identification/identify_rom_flight_correction.py` (steps 1–2, writes the ROM file); the validator with `--romaerostructural_flight--steeringsolved--out` on training cycles 20–27, 40–47 and 75–82, into `results/LEI-V3-KITE/identification/rom_flight_correction/output_error/train_*.csv`; then `refine_rom_flight_output_error.py` (step 3, the turn-rate-law gain by default, edits the ROM file in place), the held-out validation in both modes and `plot_rom_turn_rate_law.py`; the same with `--romsemi_empirical` (training CSVs under `output_error_semi_empirical/`) recalibrates the paper's ROM;
6. `scripts/identification/report_rom_aerostructural.py`, then `latexmk-pdf` in `docs/identification/`.

References

- [1] O. Cayon, V. van Deursen and R. Schmehl, *Translational dynamics of bridled kites: a reduced-order model in the course reference frame*, Wind Energ. Sci. 11, 1097–1121, 2026.
- [2] O. Cayon, M. Gaunaa and R. Schmehl, *Fast aero-structural model of a leading-edge inflatable kite*, Energies 16, 3061, 2023.
- [3] J. A. W. Poland, J. M. van Spronsen, M. Gaunaa and R. Schmehl, *Wind tunnel load measurements of a leading-edge inflatable kite rigid scale model*, Wind Energ. Sci. Discuss., 2025.